

Why Equilibrium Statistical Mechanics Works: Universality and the Renormalization Group*

Robert W. Batterman^{†‡}

Department of Philosophy, Ohio State University

Discussions of the foundations of Classical Equilibrium Statistical Mechanics (SM) typically focus on the problem of justifying the use of a certain probability measure (the microcanonical measure) to compute average values of certain functions. One would like to be able to explain why the equilibrium behavior of a wide variety of distinct systems (different sorts of molecules interacting with different potentials) can be described by the same averaging procedure. A standard approach is to appeal to ergodic theory to justify this choice of measure. A different approach, eschewing ergodicity, was initiated by A. I. Khinchin. Both explanatory programs have been subjected to severe criticisms. This paper argues that the Khinchin type program deserves further attention in light of relatively recent results in understanding the physics of universal behavior.

1. Introduction. In the introduction to his important book *Mathematical Foundations of Statistical Mechanics*, A. I. Khinchin (1949) provides a “methodological characterization” of statistical mechanics (SM). He remarks on what many before and many since have noted: namely, that the successes of SM are due in large part to its abstraction from the details of the systems it purports to describe. The aim of SM

*Received March 1997.

[†]Send reprint requests to the author. Department of Philosophy, 350 University Hall, Ohio State University, Columbus, OH 43210.

[‡]This material is based upon work supported by the National Science Foundation under Award No. SBR-9529052. I would like to thank Roger Jones, David Malament, and Abner Shimony for helpful comments and encouragement. A version of this paper was read at the 1997 APA Central division meetings in Pittsburgh. I would especially like to thank Yuri Balashov for his insightful criticisms as commentator there. I hope I have been able to address some of his worries.

Philosophy of Science, 65 (June 1998) pp. 183–208. 0031-8248/98/6502-0001\$2.00
Copyright 1998 by the Philosophy of Science Association. All rights reserved.

is to explain the properties and behaviors of a wide class of systems (mechanical systems) by considering only the most fundamental mechanical properties common to these systems.

Those general laws of mechanics which are used in statistical mechanics are necessary for any motions of material particles, no matter what are the forces causing such motions. It is a complete abstraction from the nature of these forces, that gives to statistical mechanics its specific features and contributes to its deductions all the necessary flexibility. This is best illustrated by the obvious fact that if we modify our point of view on the nature of the particles of a certain kind of matter and on the character of their interaction, the properties of this kind of matter established by methods of statistical mechanics remain unchanged by these modifications because no special assumption was made in the process of deduction of these properties. (Khinchin 1949, 8)

Khinchin's book, in essence, attempts to provide an explanation for why the prescriptions of equilibrium SM (particularly, the so-called Gibbs phase averaging method) work for computing equilibrium values of various thermodynamical quantities. That is, he is concerned to justify the computation of phase averages using Gibbs' method in which observed values for thermodynamic quantities are computed using the microcanonical measure. (To be explained below.)

Ergodic theory has been thought by many to play a fundamental role in justifying Gibbs' method. In particular, numerous attempts both before and after Khinchin's work have tried to justify the use of the microcanonical measure for computing phase averages by appealing to the ergodic nature of the dynamical motions of the systems under consideration.

On the other hand, it is well-known that Khinchin's program for explaining the success of equilibrium SM largely avoids an appeal to ergodicity. The features of his proposal which do most of the explanatory work are (i) the special nature of the functions representing the macroscopic observables and (ii) the large number of degrees of freedom characteristic of systems exhibiting thermodynamic behavior. These two features are primarily responsible for the applicability of the formalism of the theory of probability. In particular, they allow for the use of the Central Limit Theorem (CLT) as a means for determining the dispersions of values for the macroscopic observables about their average values. Nevertheless, Khinchin's program has been criticized, primarily for failing to justify the use of the microcanonical distribution.

My concern in this paper is to re-examine the Khinchin type program for equilibrium SM. In particular, I want to ask why equilibrium

SM should work virtually independently of the details of the microstructure of the systems. What accounts for the fact, noted by Khinchin in the passage quoted earlier, that statistical mechanical arguments work *because of*, and not just in spite of, its “complete abstraction” from the nature of the forces and types of interactions responsible for the actual motions of the systems being studied? These are questions concerning the explanation of a kind of universality. One wants to know why the *method* of equilibrium SM—the Gibbs’ phase averaging method—is so broadly applicable; why, that is, do systems governed by completely different forces and composed of completely different types of molecules succumb to the same method for the calculation of their equilibrium properties. I shall propose a framework within which these explanatory why-questions may profitably be discussed.

As I see it, the ergodic accounts and the Khinchin type program both seek to address the question of why equilibrium SM works. However, they differ somewhat in how that question is to be understood. The ergodic approaches construe the question rather *narrowly*, asking for a justification of the use of the microcanonical measure for computing phase averages. On my view Khinchin’s program should be seen as providing the beginning of a answer to the question more broadly construed: Why do the prescriptions of equilibrium SM yield proper results virtually without regard for any of the microscopic details of the systems being investigated? As the argument develops, I hope this distinction will come more clearly into focus.

In the next section I shall present a few of the details of the ergodic theory “explanation” for why equilibrium SM works, and then briefly consider some powerful objections to the account. Section 3 outlines the essential features of Khinchin’s program for SM and discusses its limitations. In particular, Khinchin provides arguments that apply directly only to systems of noninteracting components; whereas, the systems treated in SM are of necessity composed of energetically interacting components. I call this the “paradox of interaction.” Section 4 focuses on two further problems with the proposal. First, thermodynamic systems can undergo phase transitions even with weakly interacting components. Second, Khinchin’s proposal does not seem to address the question that is the primary concern of the ergodic approach, namely, justifying the use of the microcanonical measure. In addition, this section presents the details of an objection to the ergodic proposal due to Earman and Rédei which has the virtue (or so I shall argue) of indicating an avenue through which one might approach the entire question of accounting for the success of equilibrium SM. This avenue is explored in the final section where I outline the renormalization group framework and the explanation of universality it offers. I argue

that this framework can fruitfully be employed to investigate the explanatory question with which we are concerned.

2. The Ergodic Proposal. The traditional approach to explaining why equilibrium SM works appeals to ergodicity. Let me briefly sketch the simplest proposal—one that can be criticized on many different fronts but which nevertheless clearly provides the main motivation for the appeal to ergodicity. To begin we need a definition of this key concept. Consider a dynamical system to be a triple (Γ, ϕ_t, μ) . Γ is a phase space—a space of possible states of the system. $\phi_t : \Gamma \rightarrow \Gamma$ is a one parameter group of automorphisms (the flow) with time t the parameter. $\phi_t(x)$ is the state of the system at time t if the state at $t = 0$ was x for $x \in \Gamma$. Finally, μ is a normalized measure on Γ that is invariant under the flow. Invariance means that for any measurable set $A \subseteq \Gamma$, $\mu(\phi_t(A)) = \mu(A)$ for all t .

One useful characterization of ergodicity is the following: A dynamical system is ergodic if and only if it is metrically transitive or metrically indecomposable. This means that it is not possible to partition the phase space Γ into two or more regions A and B of nonzero measure which are invariant under the flow. In other words, the system is metrically transitive (and hence ergodic) just in case for any two regions A, B such that $A \cap B = \emptyset$ and $A \cup B = \Gamma$ which are invariant under the flow (for all t , $\phi_t(A) \subseteq A$ and $\phi_t(B) \subseteq B$) either $\mu(A) = 0$ and $\mu(B) = 1$ or $\mu(A) = 1$ and $\mu(B) = 0$.

The so-called ergodic theorem asserts the equality of infinite time averages with phase averages if and only if the system is ergodic. More precisely, we define the infinite time average of a function $f(x)$, $\hat{f}(x)$, as follows:

$$\hat{f}(x) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{t_0}^{t_0 + T} f(\phi_t(x)) dt.$$

The phase average of $f(x)$ is given by

$$\bar{f}(x) = \int_{\Gamma} f(x) d\mu(x).$$

The ergodic theorem states the equivalence of the following two claims:

- (i) For any integrable phase function f and for almost all $x \in \Gamma$, $\hat{f}(x) = \bar{f}(x)$.
- (ii) The system (Γ, ϕ_t, μ) is ergodic.

For Hamiltonian systems the flow ϕ_t in the phase space $\Gamma = \mathfrak{R}^{2N}$ is generated by Hamilton's equations of motion. If the system is conser-

vative, the motion is confined to a $2N - 1$ dimensional hypersurface of constant energy, Γ_E . There is a normalized, ϕ_t -invariant measure which corresponds to a uniform distribution over this energy surface called the microcanonical measure. Let this measure be denoted by μ . Ergodicity guarantees that the microcanonical measure, μ , is the *unique* stationary measure on the energy surface that is absolutely continuous with the Lebesgue measure.¹

On what I'm calling the "ergodic" proposal, one assumes that equilibrium values of macroscopic thermodynamic quantities can be identified with infinite time averages of appropriate phase functions. One tries to justify this assumption by appeal to the fact that the measurement of the thermodynamic quantity will typically take a long time relative to the time scale on which the microscopic processes (e.g., collisions between molecules) are occurring; and so macroscopic measurements can be nicely approximated by infinite time averages. Since calculating the infinite time averages involves completely solving the equations of motion for an N -component system (where for a typical gas N is on the order of 10^{23}), this, by itself, is a hopeless task. On the other hand, ergodicity guarantees that such time averages $\hat{f}$ are equal almost always to microcanonical phase averages $\bar{f}$, and the latter are easy to calculate. Since ergodicity also guarantees, as we have seen, the uniqueness of the microcanonical measure μ , we have the beginning of an explanation for why the Gibbs' averaging method works. Ergodicity clearly plays a central role in this account.

As already noted, this explanation can be criticized at a number of points. First, since we do often witness systems that are not in equilibrium, it is difficult to maintain the identification of thermodynamical values with *infinite* time averages. Second, there is a serious problem with interpreting the "almost always" or "almost everywhere" qualification in the identification of $\hat{f}$ with $\bar{f}$ given ergodicity. This is, in effect, equivalent to asking why the measure μ should be taken to represent physical probability. The uniqueness of μ as an invariant measure on Γ_E takes us some way towards answering this question, but the extent to which it succeeds remains a matter of debate.²

A recent paper by Earman and Rédei (1996) continues the critique of the explanatory efficacy of ergodic theory. They too hold that er-

1. A measure, μ , is absolutely continuous (a.c.) with another μ' iff for any measurable set $A \subseteq \Gamma$, $\mu'(A) = 0$ [only if $\mu(A) = 0$]. In other words, μ' agrees with μ on assignments of zero measure to sets in Γ .

2. See Malament and Zabell 1980 for the positive argument and Sklar 1993 for a detailed critique.

godic theory is irrelevant for explaining the success of equilibrium SM. They offer two main reasons for this claim. First and foremost, they point to the fact that the systems typically treated by SM have not been demonstrated to be ergodic (Earman and Rédei 1996, 69–70). Only very idealized models of systems, e.g., an ideal gas modeled as a system of perfectly hard spheres in a box, have been proven to be ergodic. Real gas molecules do not interact as perfectly elastic spheres. As they say, “the evidence for the applicability of ergodicity where it is required is non-existent. Furthermore the evidence against the applicability is strong” (Earman and Rédei 1996, 70). This latter evidence comes from the so-called KAM theorem which leads one to expect, for molecules interacting with more realistic potentials, that the systems will not be ergodic: There will be invariant regions in the phase space (for a wide range of energies) where trajectories remain confined. The existence of these regions (called “invariant tori”) allow for the decomposition of the system’s phase space into disjoint regions in which a trajectory beginning in one such region will remain forever within that region. Such a phase space will not be metrically transitive and, hence, neither will the system be ergodic. Given this fact about most systems treated by SM, it does indeed seem like ergodicity is a red herring.

Earman and Rédei reiterate an argument of Sklar’s to express their second major complaint concerning the explanatory significance of ergodic theory. Even for systems that are ergodic, ergodicity is neither necessary nor sufficient for explaining the success of equilibrium SM. Ergodicity is not sufficient, since a system with few degrees of freedom—three hard spheres in a box—can be ergodic. But, it is quite clear that it makes no sense to speak of such a system as possessing thermodynamic properties; in particular, it makes no sense to maintain that it can be in a state of thermodynamic equilibrium. Somehow, the fact that the systems treated by SM possess large numbers of components or many degrees of freedom must play an essential role in the explanation we seek. Ergodicity is not necessary according to Earman and Rédei because they buy Sklar’s argument that there is a correct, full explanation which makes no reference whatsoever to ergodicity. Here, as do Earman and Rédei, it is best to quote Sklar himself.

Ergodic theory considers the question: Why does the natural probability distribution [the microcanonical measure] work? The answer it gives is the proven equality of phase-averages to infinite time averages. But there is a much simpler answer. And it is correct. And it is the *full* answer. And it is totally independent of any ergodic results. It goes like this: How a gas behaves over time depends on (1) its microscopic constitution; (2) the laws governing

the interaction of its micro-constituents; (3) the constraints placed upon it; (4) *the initial conditions characterizing the microstate of the gas at a given time*. (Sklar 1973, 210)

Sklar emphasizes the importance of this last clause.

The *actual* distribution of initial states is such that calculations done by the Gibbs method, with the natural probability distribution over the ensemble and the natural reduction to phase average, ‘works.’ This is a matter of *fact*, not of law. These ‘facts’ explain the success of the Gibbs method. In a clear sense they are the only legitimate explanation of its success. (Sklar 1973, 210)

So the explanatory value of ergodicity, and ergodic theory in general, has been questioned on two fronts. First, it seems that most systems treated in equilibrium SM fail to be ergodic. This follows from the KAM theorem and is the major criticism by Earman and Rédei. Second, even if a system is ergodic, it isn’t clear to what extent that property is either necessary or sufficient for explaining the success of the Gibbs’ averaging method. This is the main thrust of Sklar’s critique.

On the one hand, these criticisms both seem to suggest that one should try to explain the success of equilibrium SM without appeal to ergodicity. In this respect they accord with Khinchin’s exploration of “[t]he possibility of a formulation without the use of metric indecomposability [i.e., metric transitivity or ergodicity]” (Khinchin 1949, 62). On the other hand, Sklar’s suggestion that the proper explanation of why the Gibbs method works appeals directly to the microscopic constitution of the system, the nature of its interactions, and the actual distribution of its initial conditions. In this respect it appears totally at odds with Khinchin’s claim that SM works in large part exactly *because of* its abstraction from these details. Sklar’s suggestion about what provides the full and correct explanation for why the Gibbs method works clearly has allegiance to a reductionist, perhaps, Deductive-Nomological (D-N) approach to explanation: A proper explanation of thermodynamic, macroscopic behavior will involve appeal to the exact nature of the system’s microscopic constitution and the laws governing its evolution determined by the nature of the forces of interaction among the microconstituents, etc. Surely this is at odds with Khinchin’s point of view.

In what follows, I would like to explore the possibility that a different kind of explanatory framework is required to account for the success of equilibrium SM. In particular, I want to address the questions raised earlier in the introduction: What explains the large degree of abstraction from the details to which Khinchin refers. As already noted, this is a request for an explanation of a kind of universal be-

havior. Khinchin shows how the CLT can be used to calculate dispersions of phase functions about their microcanonical averages. That is, he uses the Gaussian distribution, which is the limiting distribution in the CLT to provide asymptotic estimates for the values of the thermodynamic quantities (as the number of components of the systems gets large). The explanatory question we want to answer is why this should work for such a wide variety of systems.

3. Khinchin's Proposal. Khinchin proposes to reformulate the problem of justifying the identification of infinite time averages with phase averages in the language of probability theory. Suppose, for the phase function of interest, f , that its value on the energy surface differs very little from its average value $\bar{f}$ —suppose, that is, that it is a nearly constant function on Γ_E . More precisely, suppose that the phase dispersion of f relative to the microcanonical measure is small, $(f - \bar{f})^2 \leq \varepsilon$, ε small. Given this, it follows that the phase dispersion of the time average, $\hat{f}$, will be at least as small:³

$$\overline{(\hat{f} - \bar{f})^2} \leq \overline{(f - \bar{f})^2}. \quad (1)$$

The idea then is to employ the CLT to show that as the number of components of the system gets large, $\overline{(f - \bar{f})^2} \rightarrow 0$. Hence, asymptotically we see that for these functions f , the probability goes to zero that the time average differs from the phase average by any specified amount.

What allows one to employ the CLT in this manner? Here we need to briefly look at what the theorem says. The CLT is a statement about the limiting behavior of the distribution function for sums of random variables as the number of random variables in the sums tends to infinity. In its simplest form we assume that the individual random variables (S_i) are independent and identically distributed. We are interested in the distribution of the sum $S(n) = \sum_{i=1}^n S_i$. The CLT states that the

3. Equation (1) says that the phase dispersion of the time average $\hat{f}$ cannot be greater than the phase dispersion of f itself. Equality holds if and only if f is a constant function. A simple proof of this inequality is the following (Truesdell 1961, 47): For h any summable function, the Schwartz inequality gives $(\bar{h})^2 \leq \bar{h^2}$. Now take the phase average of both sides: $(\bar{\hat{h}})^2 \leq \bar{\hat{h^2}}$. By the Birkhoff ergodic theorem $\bar{\hat{h^2}} = \bar{h^2}$, and so, $(\bar{\hat{h}})^2 \leq \bar{h^2}$. Now let $h = f - \bar{f}$; this yields (1).

$$\lim_{n \rightarrow \infty} P \left(\frac{S(n)}{\sigma \sqrt{n}} < x \right) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt.^4$$

That is, the probability that the normalized sum $S(n)/\sigma\sqrt{n}$ has a value less than x converges, as $n \rightarrow \infty$, to the Gaussian or normal distribution. The normalization factor is clearly proportional to $\sqrt{n}$ which expresses the square root law of fluctuations. This means that the typical effect of several random contributions to the sum is of the order of $\sqrt{n}$. Since $\sqrt{n}$ increases more slowly than n , this tells us that the effect of the random contributions to the collective behavior increases much more slowly than does the number of terms in the sum.

Khinchin is able to apply this theorem for the estimation of phase dispersions of functions f representing thermodynamic quantities in part because he assumes that these f 's have a particular structure. They are so-called "sum functions." They have essentially the same form as the sum $S(n)$. As he says, the theorem will apply in part because of "the peculiar properties of mechanical systems treated in statistical physics (breaking up into a large number of components), and partially [because of] the specific properties of the functions with which we are dealing (these are, as a rule, the 'sum-functions', i.e., the sums of functions each depending on the dynamical coordinates of only one component" (Khinchin 1949, 63). In fact, that the functions he considers are sum functions is responsible for his being able to assume their near constancy on the energy surface in the first place. This is an expression of the law of large numbers.

It is important to understand exactly how restrictive Khinchin's proposal is. By abandoning the goal of showing that a *system*, (Γ, ϕ, μ) , is ergodic, Khinchin gives up on showing that time averages equal phase averages for almost all phase functions f . Instead, his aim is to argue that certain special *functions*—sum functions which presumably represent macroscopic or thermodynamic quantities—are "ergodic." That is, that the time averages for these special functions are nearly equal to their phase averages.⁵

Khinchin 1949, Section 23 and Chapter 6, treats in some detail the example of a monatomic ideal gas. This is a system composed of a

4. Note that here we have assumed that the expectations of $S_i, \bar{S}_i$ equal zero for all i .

5. There is a serious worry really about how realistic the restriction to sum functions is. Despite what Khinchin says, many functions of interest in statistical mechanics do not have this special form. At best we must take sum functions to be a proper subclass of functions which exhibit the appropriate "nearly constant" behavior on the energy surface; and hence the explanatory program outlined here would need to be extended to the full class of functions of interest.

large number of molecules that are treated as point particles. The total energy of this system is simply the sum of the kinetic energies of the individual molecules. That is, there is no mutual potential energy or interaction energy.⁶ The energy, therefore, is a sum function.

An important quantity is the so-called structure function for the system. For a system with Hamiltonian $H(x)$ the structure function is given by:

$$\Omega(x) = \int_{\Gamma} \delta(E - H) dx_{2N}.$$

It is the “volume” of the surface of constant energy with respect to the Lebesgue measure, and plays the role of the normalizing factor in the definition of the microcanonical distribution:

$$\mu(x) = \frac{\delta(E - H)}{\int_{\Gamma} \delta(E - H) dx_{2N}}.$$

Therefore, the structure function clearly plays a role in determining the average value of any arbitrary phase function on the energy surface Γ_E :

$$\bar{f}(x) = \int_{\Gamma} f(x) d\mu(x).$$

Khinchin is able to show that the probability distribution for the energy of a given component of a system is determined by the structure functions of that component itself, the structure functions for the other components, and the structure function for the entire system. He shows how one can find approximate expressions for the structure function of systems like the ideal monatomic gas that are composed of a large number of similar components. (The components will all have structure functions of the same basic form.) As he says, “[u]sing the methods of the theory of probability we will be able to establish for the structure functions of such [large] systems the approximate expressions which are to a large extent independent of the nature of individual components” (Khinchin 1949, 75).

In effect Khinchin shows how one can treat the energy, say, of each of the components of the ideal gas as independent and identically distributed random variables. Their common structure functions determine their common distribution function. One can then employ the CLT to determine the asymptotic value for the energy of a large component of the system—a component composed of many molecules. The result, of course, is that the energy will be Gaussian distributed and strongly

6. The importance and ramifications of this idealization will be discussed below.

peaked about the mean with root-mean-square deviation proportional to the square root of the number of components, N , as $N \rightarrow \infty$.

The entire argument depends on the possibility of treating a large system as being decomposable into components. Furthermore, it is necessary that the phase functions representing the thermodynamic quantities for the entire system be sums of phase functions of these components. For example, if the energy $E = E(x_1, \dots, x_{2N})$ can be written as the sum $E = E_1(x_1, \dots, x_i) + E_2(x_{i+1}, \dots, x_{2N})$ we say that the system can be decomposed into two components represented by the coordinates $(x_1, \dots, x_i)$ and $(x_{i+1}, \dots, x_{2N})$. These components each have their own phase spaces Γ_1 and Γ_2 whose direct product is the phase space Γ for the whole system. Likewise each component has a structure function Ω_1 and Ω_2 . The structure function for the entire system is the convolution of the structure functions of its components:

$$\Omega(x) = \int_0^\infty \Omega_1(y)\Omega_2(x - y)dy.$$

Of course, for an n component system, the structure function is the n -fold convolution of the structure functions of the components (Khinchin 1949, 41).

This assumption of decomposability is essential for the program. But it brings with it what Khinchin (1949, 41) takes to be a “methodological paradox.” We can call this the “paradox of interaction.” The decomposability of the system into components in the sense just described excludes the possibility that the components interact with one another energetically. As Khinchin says,

[i]ndeed, if the Hamiltonian function, which expresses the energy of our system, is a sum of functions each depending only on the dynamic coordinates of a single particle (and representing the Hamiltonian function of this particle), then, clearly, [Hamilton’s system] of equations . . . splits into component systems [of equations] each of which describes the motion of some separate particle and is not connected in any way with other particles. Hence the energy of each particle, which is expressed by its Hamiltonian function, appears as an integral of equations of motion, and therefore remains constant. (Khinchin 1949, 42)

The paradox arises because it is a presupposition of the applicability of SM that the particles (say, the molecules of a gas) are “in a state of intensive energy interaction, where the energy of one particle is transferred to another (for instance by means of collisions)” (Khinchin 1949, 41–42).

Khinchin's response to the paradox of interaction is to state that we must really think of the particles as "only approximately isolated energetically components." He holds that we must, when being precise, allow for correlations between components which would strictly speaking block the kind of decomposition into individual components we have been considering. He says,

inasmuch as forces of interaction between the particles manifest themselves only at very small distances, such mixed terms in the expression of energy, representing mutual potential energy of particles, will be (in the great majority of points of the phase space) negligible as compared with the kinetic energy of particles or with the potential energy of external fields. In particular they will contribute very little in evaluating various averages. . . . However, these mixed terms that are neglected, from the point of principle play a very important role, since it is precisely their presence that assures the possibility of an exchange of energy between the particles, on which is based the whole of statistical mechanics (Khinchin 1949, 42–43)

The paradox of interaction is clearly a concern. And, Khinchin's rather handwaving response surely requires deeper justification. Two responses to this problem immediately present themselves. First, one might try to make precise Khinchin's vague claim that the "mixed terms" representing the mutual potential energy will be negligible in comparison with the terms for the kinetic energy of the components and the energies of external fields. C. Truesdell (1961, 55) formulates this as a program for further study: Khinchin's argument has depended on the separability of the Hamiltonian; namely, that $H(x) = \sum_{i=1}^N H_i(x_i)$. For a separable Hamiltonian, though, for each component, H_i we know that $H_i(x_i) = \text{constant}$ is an integral of the motion, and so there is no interaction at all. Truesdell sees Khinchin as imagining that the Hamiltonian for the real system is best expressed in the following form:

$$H(x) = \sum_{i=1}^N H_i(x_i) + \delta H_{int}(x_1, x_2, \dots, x_N),$$

and allowing $\delta \rightarrow 0$. That is, according to Truesdell, Khinchin's results can be represented as holding in the following limit:

$$\lim_{N \rightarrow \infty} \lim_{\delta \rightarrow 0}$$

But, Truesdell argues that “physically adequate theorems should refer to the inverted limit”:⁷

$$\lim_{\delta \rightarrow 0} \lim_{N \rightarrow \infty}.$$

While this formalizes Khinchin’s handwaving, as far as I know, there are no exact results referring to the “inverted limit.” Furthermore, while I believe it is clear that the inverted order of the limits is more realistic, it is not so obvious (to me at least) that there is a *physical* justification for letting $\delta \rightarrow 0$ *after* taking the limit of large systems.

The second approach to the paradox of interaction is to keep $\delta > 0$ and see if one can still show that the phase dispersion of the appropriate phase functions gets small as the number of components of the system gets large. This is the aim of the so-called “Theory of the Thermodynamic Limit.” In other words, the goal is to try to prove limit theorems for systems with interacting components as $N \rightarrow \infty$, and the volume $V \rightarrow \infty$ but where the density N/V remains constant. This program has been quite successful. Work by Ruelle (1969), Lanford (1973), and others have led to rigorous theorems demonstrating the existence of the thermodynamic limit for systems with more realistic interaction potentials. The potentials for which such limits exist must satisfy so-called stability and tempering conditions. The stability condition demands that the potential be bounded “from below.” Crudely speaking, this assures that an infinite number of particles will not “collapse” into some bounded spatial region. The tempering condition guarantees that the strength of the interaction between particles falls off sufficiently as their separation increases.

Results such as these lead one to expect that even for systems with interacting components we can expect Khinchin-type “Central Limiting behavior”: For the right sort of phase functions, their values on a surface of constant energy will be peaked around the most probable value with narrow dispersions determined by the Gaussian law—that is, with their dispersions asymptotically proportional to the number of components. Mazur and van der Linden (1963), in fact, explicitly demonstrate that Khinchin’s asymptotic formula for the structure function of a system of noninteracting components, also holds for systems interacting with more realistic potentials.

4. Problems with Khinchin’s Program. There are two main problems with the program outlined in the last section. First, the limit theorems

7. See Truesdell 1961, 55. I have corrected a missprint in the specification of the order of these limits in Truesdell’s paper.

can fail, even for systems with weakly interacting components if the system is undergoing a phase transition. Even “small” interaction terms can combine to realize large effects when a system is at a critical point. This fact is, of course, explicitly recognized by those people investigating the thermodynamic limit. Nevertheless, it would be nice to have a unified account of large systems encompassing systems both near and far from their critical points. I shall have more to say about such a framework in the next section.

Second, there is a sense in which the Khinchin type programs completely fail to answer the main question to which the ergodic proposal addressed itself. This failing has been explicitly recognized by Truesdell and Sklar. For instance, Sklar (1993, 163) notes that “the Khinchin theorem only tells us that there is a ‘high probability asymptotically going to one’ that a system in a given micro-state will have its infinite time average of the appropriate phase function equal to the phase average of that function.” Furthermore, he argues that demonstrating this probability one claim is not sufficient. After all, these probabilities are themselves being computed with respect to the microcanonical measure. But it is the justification of the use of the microcanonical measure which is exactly at issue. As we saw earlier in Section 2, ergodicity is invoked in the attempt to justify the use of this measure for computing phase averages. But on the current proposal, we are trying to do without having to prove the system to be ergodic.

Suppose that we have failed to notice some global constant of motion other than the energy; e.g., suppose angular momentum is also a conserved quantity. This means that the system’s actual state is confined to a subspace of the energy surface of dimension $2N - 2$. The Khinchin type results show that, asymptotically, there is a zero probability that the time average of f differs from the phase average of f on the $2N - 1$ dimensional energy surface. But, relative to the microcanonical measure on the energy surface, the subspace to which the system is actually confined has measure zero. In this situation, the result of calculating phase averages with respect to the microcanonical measure on the full energy surface and identifying these with the values for thermodynamic quantities will generally yield completely erroneous results. Of course, if the system is ergodic, then we know there are no global constants of motion that we have missed. But, the entire point of Khinchin’s program is to make an “end run” around having to demonstrate ergodicity. It looks like the move to focus on the large number of components of the systems treated by SM, and the special nature of the phase functions typically representing thermodynamic quantities has not helped with the fundamental question addressed by the ergodic proposal outlined in Section 2.

Faced with the fact that most systems treated by SM are not ergodic, as the KAM theorem demonstrates, Wightman has made the following proposal. Given the applicability of the KAM theorem we have a system whose energy surface Γ_E can be decomposed into at least two invariant regions of nonzero Lebesgue measure, one of which is ergodic and the other not. Wightman (1983, 20) suggests that perhaps, as N and V get large, the relevant thermodynamic observables are represented by phase functions that are “insensitive to the non-ergodic portion of the flow even if its relative phase volume does not go to zero.” The focus on large N and V , as well as on particular phase functions exhibits a clear affinity with the Khinchin type, thermodynamic limit proposals.⁸

However, Earman and Rédei raise the following objection to Wightman’s proposal. Let us call the ergodic region A and the nonergodic “KAM” region B . In this situation there are many normalized ϕ_t -invariant measures that are a.c. with respect to the Lebesgue measure other than the microcanonical measure μ . Earman and Rédei construct the following family of measures:

$$\mu_\varepsilon = \varepsilon\mu_A + (1 - \varepsilon)\mu'_B \quad (0 \leq \varepsilon \leq 1). \quad (2)$$

Here “ μ_A ” is the unique (since the flow on A is ergodic) normed invariant measure on A a.c. with respect to the Lebesgue measure defined as follows:⁹

$$\mu_A(\bullet) = \frac{\mu(\bullet \cap A)}{\mu(A)}.$$

And, μ'_B is any normed, a.c., invariant measure whose support is B . By construction, for any ε , μ_ε is a normed, ϕ_t -invariant, a.c. measure on the entire energy surface Γ_E .

If we compute the phase average of f with respect to μ_ε we get the following:

$$\bar{f}_\varepsilon = \frac{\varepsilon}{\mu(A)} \int_{\Gamma_E} f(x) d\mu(x) + (1 - \varepsilon) \int_{\Gamma_E} f(x) d\mu'_B(x)$$

(Recall that μ is the microcanonical measure.)

The objection is that by adjusting ε one can make the phase average of the observable represented by f as insensitive to the ergodic portion of the flow (that is, region A) as one would like. In other words, if ε is

8. I will suggest below that what one needs to show is that the *calculated averages* of the functions are insensitive to the nonergodic portion of the flow.

9. See Earman and Rédei 1996, 72.

small, then $\tilde{f}_\varepsilon$ is dominated by the second term in the equation above. In such a situation, it is surely to be expected that $\tilde{f}_\varepsilon$ will yield incorrect predictions for the actual value of the thermodynamic observable represented by the function f . But, as Earman and Rédei (1996, 72) note, “ergodic theory does not explain why this [is to be expected].” It seems to me that we must grant Earman and Rédei’s conclusion. Ergodic theory does *not* explain why we should expect $\tilde{f}_\varepsilon$ to yield poor values for the thermodynamic quantities when ε is sufficiently small. Nevertheless, I think that there is some deep merit to Wightman’s proposal not considered by Earman and Rédei. In the next section, I shall describe a framework that offers us the promise of determining for which values of ε we can expect averages with respect to μ_ε to yield reasonable results. The idea is to ask to what extent ε may be diminished and yet in the limit of large N and V , we can still expect the Gaussian distribution to emerge. In effect, what we are asking for is an account of the robustness or universality of the Gaussian distribution or, equivalently, of Central Limiting behavior.

5. The Renormalization Group. In the last section, I noted that the limit theorems established in the rigorous theory of the thermodynamic limit, while extending the results of Khinchin, do not hold when the system is at a critical point. Systems undergoing phase transitions require different treatment. At the critical point, components of the system that are widely separated spatially become strongly correlated. In the language of the thermodynamic limit, this would mean that the tempering condition fails to obtain for such systems.

It turns out that systems undergoing phase transitions typically exhibit the following property: Their behavior at the critical point is characterized by a small set of dimensionless numbers called “critical exponents.” What is truly remarkable about these numbers is their universality: It is a well-established experimental fact that systems as diverse as different fluids composed of different kinds of molecules with different forces of interaction between them, and even lattice systems such as ferromagnets, have critical behavior characterized by the *same* critical exponents. In other words, the critical behavior of systems whose components and interactions are radically different is virtually identical. Hence, such behavior must be largely independent of the details of the microstructures of the various systems. This is known in the literature as the “universality of critical phenomena.”

Surely one would like to account for this universality. The so-called “renormalization group” provides the desired explanation. Let me give a broad outline of the form this explanation takes.

Every system, say each different fluid, is represented by a function—its Hamiltonian. The Hamiltonian characterizes the kinds of interactions between the system's components (or degrees of freedom), the effects of external fields acting upon the system, etc. When the system is not in the critical regime, its different components are correlated (because of interactions) only weakly with one another. However, when a system is near its critical point the “length” of the range of correlations between the different degrees of freedom increases. In fact, at the critical point, this so-called “correlation length” diverges to infinity. The divergence of the correlation length is intimately associated with the system's critical behavior. It means that correlations at every length scale (between near as well as extremely distant components) contribute to the physics of the system at its critical point. In effect, this constitutes a highly singular mathematical problem, one which is completely intractable. While it is relatively easy to deal exactly with correlations that obtain between *pairs* of particles, the situation is completely hopeless when one must consider correlations between three or more—indeed more than 10^{23} —particles. The guiding idea of the renormalization group method is to find a way of turning this singular problem into one which is regular and tractable. This miraculous alchemy is effected by transforming the problem into one of analyzing the topological structure of an appropriate abstract space—the space of Hamiltonians.

One introduces a transformation on this space that maps an initial physical Hamiltonian describing a real system to another Hamiltonian in the space. The transformation preserves, to some extent, the form of the original physical Hamiltonian so that when the interaction terms are properly adjusted (renormalized), the new renormalized Hamiltonian describes a system exhibiting the same or similar thermodynamical behavior. Most importantly, however, the transformation effects a reduction in the number of coupled components or degrees of freedom within the correlation length. Thus, the new renormalized Hamiltonian describes a system which presents a more tractable problem. It is to be hoped that by repeated application of this renormalization group transformation the problem becomes more and more tractable until one can solve the problem by relatively simple methods. *In effect, the renormalization group transformation eliminates those degrees of freedom (those microscopic details) which are inessential or irrelevant for characterizing the system's dominant behavior at criticality.*

In fact, if the initial Hamiltonian describes a system at criticality, then each renormalized Hamiltonian must also be at criticality. The sequence of Hamiltonians thus generated defines a trajectory in the abstract space that, in the limit as the number of transformations goes

to infinity, ends at a fixed point. The behavior of trajectories in the neighborhood of the fixed point can be determined by an analysis of the stability properties of the fixed point. *This analysis also allows for the calculation of the critical exponents characterizing the critical behavior of the system.* It turns out that *different physical* Hamiltonians can flow to the same fixed point. Thus, their critical behaviors are characterized by the same critical exponents. This is the essence of the explanation for the universality of critical behavior: Hamiltonians describing different physical systems fall into the basin of attraction of the same renormalization group fixed point. This means that if one were to alter, even quite considerably, some of the basic features of a system (say from those of a fluid F to fluid F' composed of a different kind of molecule and a different interaction potential), the resulting system (F') will exhibit the same critical behavior. This *stability* under perturbation demonstrates that certain facts about the microconstituents of the systems are individually largely irrelevant for the systems' behaviors at criticality. Instead, their *collective properties* dominate their critical behavior, and these collective properties are characterized by the fixed points of the renormalization group transformation (and the local properties of the renormalization group flow in the neighborhood of those points). We can, through this topological analysis, understand both how universality arises and why the diverse systems display identical behavior.

Before returning to the problem of explaining the success of equilibrium SM, it is important to realize that this type of explanation for universal behavior is ubiquitous.¹⁰ Consider the following simple, yet illustrative, example. An engineer is designing a structure such as a bridge. The work is necessarily qualitative in the following sense. Because of the extreme complexity of the materials, the engineer does not really know the exact nature of the equation that governs the behavior of the bridge. Nevertheless, she would like to be assured that the bridge is not going to collapse. Had she the exact equation, she could in principle determine the "location" of any singularities in the equation which may indicate configurations of materials for which a collapse might be possible. However, since the engineer does not have this exact equation, she is actually concerned with the equation's (structural) stability—roughly, the stability of the topology of its solutions under perturbation of its form. In other words, since the equation she works with (her dynamical system) is most likely an approximation to the actual equation, it is important to know the extent to which the two dynamical systems have the same singularity structure or will exhibit

10. For a detailed discussion see Batterman 1998.

the same qualitative behavior. The engineer needs to know that the qualitative conclusions she makes (concerning, for instance, the possibility that the bridge will collapse) on the basis of her dynamical system, will also hold for the actual dynamical system (differential equation) describing the real structure in the world. The stability analysis that provides such information is analogous to that involved in determining the fixed point structure of the space of Hamiltonians just outlined. We get an account of stability of the critical behavior under perturbation of the details of the interactions (Hamiltonians) characteristic of the systems.

It seems to me that this type of analysis, broadly conceived, does lie behind the explanation and understanding of various instances of universality discussed in the physics and mathematics literature.¹¹ My goal now is to show that the same strategy can be employed to explain the success of the Gibbs method in equilibrium SM. As we have seen, such an explanation requires that we explain the ubiquity of the Gaussian distribution as the limiting distribution for sum functions representing thermodynamic quantities of interest. My suggestion is that the relevant explanatory framework is exactly that of the renormalization group method. By translating the problem into the renormalization group framework, we may, in principle at least, have a way of rationalizing Wightman's suggestion concerning the justification of equilibrium SM without falling prey to the Earman-Rédei objection. In other words, this approach aims to respond to the objections to the Khinchin program in the spirit of the original proposal—without having to appeal to ergodicity.

A number of papers (Jona-Lasinio 1975; Cassandro and Jona-Lasinio 1978; Gallavotti and Martin-Löf 1975; Bleher and Sinai 1973, 1978, 1982) have developed a rigorous connection between renormalization group methods dealing with phase transitions and certain aspects of limit theorems in probability theory. For our purposes it suffices to establish the connection only at the most qualitative level.

To this end, let us rewrite the CLT in a slightly different form than that presented above in Section 3. Let us assume here that S_i can take the value 1 or 0 if the i^{th} toss of a coin is, respectively, heads or tails with probabilities p and $1 - p$. Recall that $S(n) = \sum_{i=1}^n S_i$. It is the number of heads in n tosses. Then the CLT says

11. Goldenfeld et al. (1989) have argued that renormalization group type analyses are, in fact, necessary for explaining and understanding virtually any macroscopic phenomenology. Barenblatt's (1996) work on so-called "intermediate asymptotics" takes a similar line.

$$\lim_{n \rightarrow \infty} P \left(a \leq \frac{S(n) - np}{\sqrt{np(1-p)}} < b \right) = \frac{1}{\sqrt{2\pi}} \int_a^b e^{-t^2/2} dt. \quad (3)$$

This describes the limiting behavior for sums of *independent* and identically distributed random variables.

We are now concerned with the possibility of limit distributions for sum functions representing thermodynamic quantities in which the summands are functions representing interacting components of realistic systems. We may still treat the component functions for large systems as random variables; because of the interactions, however, it is surely not the case that they can be treated as independent.

The least severe relaxation of the criterion of independence is to assume correlations realized by a simple Markov process. In other words, imagine we are dealing with a sequence of coin tosses in which the probability of a particular outcome on a given trial depends on the result of the previous trial. We can express this dependence in terms of a matrix of transition probabilities:

$$\begin{pmatrix} \alpha & 1 - \alpha \\ \beta & 1 - \beta \end{pmatrix}$$

Say a head occurs on the i^{th} trial. Then the first row states that on the $(i + 1)^{\text{st}}$ trial, the probability of heads is α and the probability of tails is $1 - \alpha$. According to the second row, if tails occurs on the i^{th} trial, then the probability of heads on the $(i + 1)^{\text{st}}$ trial is β and the probability of tails is $1 - \beta$. Let p be the probability of heads on the first trial and assume that $0 < \alpha, \beta < 1$. Given all of this, it is possible to prove the following limit theorem:

$$\begin{aligned} \lim_{n \rightarrow \infty} P \left(a \leq \frac{S(n) - np}{\sqrt{np(1-p) \frac{(1 + \alpha - \beta)}{(1 - \alpha + \beta)}}} < b \right) &= \frac{1}{\sqrt{2\pi}} \int_a^b e^{-t^2/2} dt. \end{aligned} \quad (4)$$

The important feature to note here is that the *same* asymptotic regularity is obtained in the case where there is this Markov dependence or correlation between the trials. (See Jona-Lasinio 1975, 101–102). That is, the right hand sides of the two limit theorems (3) and (4) are identical. The collective behaviors are the same. From this example we learn that there are at least some distributions for sums of *dependent or correlated* random variables that converge to the Gaussian distribution in the limit of a large number of trials.

We also know that this Central Limiting behavior fails in the case of phase transitions and critical behavior where the correlation or dependence is in some sense strong. In such a case the probabilistic interpretation of the renormalization group treats the fixed points of the renormalization group transformation as non-Gaussian limit distributions for sums of *strongly dependent* random variables.¹² The universality of critical behavior is explained through the determination of the basins of attraction of these non-Gaussian fixed points.

The suggestion now is that we explain the universality of the *Gaussian* distribution in a completely analogous way (see Sinai 1992, Chapter 15 for a nice discussion of the strategy). Furthermore, since having the Gaussian distribution emerge in the limit as the number of components of the system goes to infinity is apparently essential for realizing Khinchin's nonergodic justification of Gibbs' method, we may thereby explain the success of equilibrium SM. The goal, therefore, is to determine the basin of attraction of the *Gaussian* fixed point. As the simple examples of the two coin tosses shows, this domain surely can contain distributions that are weakly dependent in an appropriate sense.

But what about Wightman's proposal and the objection raised by Earman and Rédei? Wightman suggests that at least for some systems to which the KAM theorem applies, that the thermodynamic observables will be insensitive to the nonergodic portion of the phase flow even if the relative measure of that nonergodic portion does not approach zero as the system gets large (that is, as $N \rightarrow \infty$ and $V \rightarrow \infty$). Let $f(x) = \sum_i f_i(x)$ be a sum function representing a thermodynamic quantity of interest on our system. Suppose that the individual components, f_i , treated as random variables, are identically distributed with respect to the measure $\mu_\varepsilon(x)$ for some given ε . (μ_ε is defined according to equation (2) above.) To counter Earman's and Rédei's objection one would need to show that for the common distribution $\mu_\varepsilon(x)$, the distribution function for the sums $f(N) = \frac{\sum_{i=1}^N f_i}{B_N} - A_N$ (for suitable normalizations B_N and displacements A_N) converges as $N \rightarrow \infty$ to the Gaussian.

In the renormalization group framework we are now considering, this amounts to showing that the distribution μ_ε is in the domain of

12. Sinai 1982 has rigorously constructed non-Gaussian limit distributions for block variables (sum functions) for certain lattice systems. The determination of these fixed points involves the so-called ε -expansion which plays such a fundamental role in the theory of Wilson (see Wilson and Kogut 1974). The " ε " in the expansion is, of course, distinct from the " ε " in Earman's and Rédei's example.

attraction of the Gaussian distribution. Determining the necessary and sufficient conditions for whether a distribution is in this domain of attraction is a completely solved problem in probability theory (at least for the case where independence is assumed)¹³ (see Jona-Lasinio 1975 and Gnedenko and Kolmogorov 1968, 171ff.). If upon investigation for real systems of interest, it turns out that for some range of values for $\varepsilon < 1$, the measures μ_ε are in the domain of attraction of the Gaussian distribution, then Wightman's proposal will have been vindicated (at least for those systems and that range of ε values).

Would this provide an explanation for why Gibb's phase averaging works? I want to claim that the answer is a qualified "yes." It can be argued, I believe, that the topological analysis involved in the renormalization group provides an acceptable explanation for the universality of critical phenomena.¹⁴ One understands why some one critical exponent characterizes the critical behavior of a vast range of distinct systems in terms of the stability of that behavior under a kind of perturbation of the microscopic details—kinds of particles, types of interactions, etc. This stability, in effect, is represented by the "size" of the domain of attraction of the relevant fixed point of the renormalization group transformation. The renormalization group provides principled reasons for why and to what extent the details of the microconstituents can be ignored.

This sort of explanation may be available in the current context, if the following conjecture can be substantiated: We can treat the parameter ε as characterizing (in a sense to be explained) the degree to which certain dynamical details are relevant to the calculation of the system's equilibrium behavior. For small values of ε in Earman and Rédei's measure, we know that the structure of invariant KAM tori on the phase space matters more and more to the averages calculated with respect to μ_ε . We can think of these tori as responsible for (or at least as descriptive of) correlations among various degrees of freedom of the system. For instance, there is surely strong correlation among degrees of freedom in the invariant KAM regions—the invariant tori. These strong dependencies guarantee the presence of weaker correlations between degrees of freedom in the assumed ergodic regions complementary to the invariant tori on the energy surface: In these regions there is at least weak correlation reflecting the fact that allowed microstates must avoid the KAM tori. Given this it seems that the μ_ε distribution of one "component" can be *strongly dependent* on the distribution of

13. Many other results are also known for cases involving dependent random variables. See, for example, Ibraghimov and Linnik 1969.

14. See Batterman 1998 for a discussion of this sort of explanation.

another “component” of the system.¹⁵ The conjecture is that ε can be plausibly related to an appropriate conception of strength of dependence or of correlation among the relevant components of the system.¹⁶

The conception of “strength” I have in mind here is the following. It is the *weight* given to the correlations resulting from the KAM tori in the calculation of average values. One can think of the measures μ_ε , for various values of ε , as specifying how much *weight* different regions on the energy surface receive in the calculation of averages. In other words, the different ε values specify the extent to which the strongly correlated degrees of freedom in the KAM tori contribute to the average value of the appropriate (sum) function representing the thermodynamic equilibrium quantity. For distributions μ_ε with ε small, the “bad” nonergodic regions will dominate the calculated averages more, and one should expect that Central Limiting behavior will fail— μ_ε will not be in the domain of attraction of the Gaussian fixed point. On the other hand, if the distribution is in that domain (if ε is sufficiently large), then *one might just as well compute phase averages with respect to the microcanonical measure μ* , since the limiting distribution for $N \rightarrow \infty$ will be the same, viz., the Gaussian. The aim is to demonstrate the robustness of the Gaussian distribution as the limit distribution for sum functions under a variation of the *importance or weight* of the correlations determined by the existence of the nonergodic regions on the energy surface.

Let me put the point another way. We do not seek to justify the microcanonical distribution as the unique appropriate distribution for calculating phase averages. Instead, our goal is to show that if the components of the system are actually weighted according to some measure $\mu_\varepsilon \neq \mu$ in the domain of attraction of the Gaussian distribution, then in the limit of large numbers of components the asymptotic formulas for the thermodynamical sum functions will yield results as *if* the components were weighted according to the microcanonical measure. But of course, many probability measures will be justified on this

15. On Khinchin’s proposal (1949, 38–39), a component of a system need not be a physically distinct entity (like an individual molecule)—the set of values of a degree of freedom or the phase space subspace representing that set of values counts as a component.

16. In the discussions of the probabilistic version of the renormalization group, strong dependence is explicated in terms of a failure of so-called “strong mixing,” where for lattice systems strong mixing holds if one cannot compensate for weakening dependence between components by increasing the spatial distance between them. In ergodic theory, on the other hand, mixing typically refers to the emergence of, or approach to, probabilistic independence in the infinite time limit. I am unsure exactly of what connections, if any can be made between these two notions.

proposal. There is really nothing very special about the microcanonical measure.

6. Conclusion. A number of investigators have asked why equilibrium SM works. But what, exactly, is the explanandum? Most interpreters have understood the question rather *narrowly*, as a demand for justifying the computation of phase averages using the microcanonical measure. The majority of responses have involved appeals to ergodicity and other properties in the ergodic hierarchy. As we have seen, though, such appeals are unlikely to work—at least for the majority of systems to which equilibrium SM is supposed to apply.

Khinchin's program is a notably different. He tries to show that an appeal to the nature of the sorts of phase functions that typically represent the thermodynamic quantities of interest (their being sum functions), together with the fact that such systems are composed of an extremely large number of similar components, suffices to justify the use of Gibbs method in equilibrium SM. This program has been extended and continued in the investigations of the thermodynamic limit by Ruelle, Lanford, and others. As I noted in Section 3, it has had many successes. In particular, what I called the paradox of interaction is dealt with quite effectively.

In spite of these successes, it seems that the objections raised in one form or another by Truesdell, Sklar, and Earman and Rédei still have some force. If the system is *not* ergodic, then it is possible to construct a.c. invariant measures of the μ_e -variety that are, *prima facie*, on an equal footing with the microcanonical measure. According to Earman and Rédei, this raises the problem of finding a rationale for choosing the microcanonical measure among all of these competitors. Their conclusion is that ergodic theory seems unable to provide such a rationale. Perhaps, in order to explain why calculating averages with respect to the microcanonical measure yields correct results, one will need to appeal explicitly, as Sklar has suggested, to the microscopic details, the nature of the molecular interactions, and the matter of fact distributions of initial conditions in the world.

However, an appeal to the detailed nature of the microconstituents and their interactions entails that we give up any attempt to answer the question of why equilibrium SM works, where that question is construed in a *broader sense*. The broad sense concerns why the prescriptions of equilibrium SM yield proper results virtually without regard for these microscopic details. That is, it is a question about the universal applicability of the method. Providing the details for each case individually does not give us any explanation for the universality which for many is an essential characteristic of SM. Recall Khinchin's

explicit claim that “[i]t is a complete abstraction from the nature of [the molecular] forces, that gives to statistical mechanics its specific features and contributes to its deductions all the necessary flexibility” (Khinchin 1949, 8).

My aim in this paper has been to set out a framework within which this broader question of explaining a form of universality can be addressed. Renormalization group analyses have been used to explain, and provide understanding of, the universality of critical phenomena (see Batterman 1998). The probabilistic formulation of the renormalization group strategy allows for a direct application of the same sort of analysis to account for the ubiquity of the Gaussian distribution for noncritical systems studied in SM. The suggestion is that we can explain why equilibrium SM works, if we can provide principled reasons for why the details of the microconstituents are by and large inessential for accounting for the thermodynamic or macroscopic behavior observed. The statistical mechanical explanation of this behavior appeals essentially to the estimates for phase dispersions guaranteed by the emergence of the Gaussian distribution in the asymptotic limit of a large number of components. In other words, we can understand why equilibrium SM works, if we can characterize the domain of applicability of the CLT. The renormalization group argument provides exactly this (at least ideally) by exhibiting the stability of the Gaussian distribution (as the limit distribution for the relevant sum functions) under perturbation of certain microscopic details.

The possibility of constructing a.c. invariant measures (the μ_e measures) which are *in fact* on an equal footing with the microcanonical distribution is, therefore, *not* a problem. On the contrary, if these distributions are in the domain of attraction of the Gaussian distribution, then their existence only serves to illustrate the extent to which the microscopic details are irrelevant or inessential to the thermodynamical equilibrium behavior we wish to explain.

REFERENCES

- Barenblatt, Grigory. I. (1996), *Scaling, Self-similarity, and Intermediate Asymptotics*. Cambridge: Cambridge University Press.
- Batterman, Robert. W. (1998), “Universality, Unification, and Understanding”, Preprint.
- Bleher, P. M. and Ya. G. Sinai (1973), “Investigation of the Critical Point in Models of the Type of Dyson’s Hierarchical Models”, *Communications in Mathematical Physics* 33: 23-42.
- Cassandro, M. and G. Jona-Lasinio (1978), “Critical Point Behaviour and Probability Theory”, *Advances in Physics* 27: 913-941.
- Earman, John. and M. Rédei (1996), “Why Ergodic Theory Does Not Explain the Success of Equilibrium Statistical Mechanics”, *British Journal of the Philosophy of Science* 47: 63-78.

- Gallavotti, Giovanni. and A. Martinlöf (1975), "Block-spin Distributions for Short-range Attractive Ising Models", *Il Nuovo Cimento* 25 B: 425-441.
- Gnedenko, Boris. V. and A. N. Kolmogorov ([1949] 1968), *Limit Distributions for Sums of Independent Random Variables*. Translated by K. L. Chung. Reading MA: Addison-Wesley.
- Goldenfeld, Nigel, O. Martin, and Y. Oono (1989), "Intermediate Asymptotics and Renormalization Group Theory", *Journal of Scientific Computing* 4: 355-372.
- Ibraghimov, Ildar. A. and Yu. V. Linnik (1969), *Independent and Stationary Sequences of Random Variables*. Groenigen: Walter Noordhoff Publishing Company.
- Jona-Lasinio, G. (1975), "The Renormalization Group: A Probabilistic View", *Il Nuovo Cimento* 26 B: 99-119.
- Khinchin, Alexander. I. (1949), *Mathematical Foundations of Statistical Mechanics*. Translated by G. Gamow. New York: Dover Publications.
- Lanford, III, Oscar E. (1973), "Entropy and Equilibrium States in Classical Statistical Mechanics", in A. Lenard (ed.), *Statistical Mechanics and Mathematical Problems* Berlin: Springer-Verlag, pp. 1-113.
- Malament, David and S. Zabell (1980), "Why Gibbs Phase Averages Work—The Role of Ergodic Theory", *Philosophy of Science* 47: 339-349.
- Mazur, Peter and J. van der Linden (1963), "Asymptotic Form of the Structure Function for Real Systems", *Journal of Mathematical Physics* 4: 271-277.
- Ruelle, David (1969), *Statistical Mechanics: Rigorous Results*. New York: W. A. Benjamin.
- Sinai, Yakov G. (1978), "Mathematical Foundations of the Renormalization Group Method in Statistical Physics", in G. Dell'Antonio, S. Doplicher, and G. Jona-Lasinio (eds.), *Mathematical Problems in Theoretical Physics*. Berlin: Springer-Verlag, pp. 303-311.
- . (1982), *Theory of Phase Transitions: Rigorous Results*. Oxford: Pergamon Press.
- . (1992), *Probability Theory: An Introductory Course*. Translated by D. Houghton. Berlin: Springer-Verlag.
- Sklar, Lawrence (1973), "Statistical Explanation and Ergodic Theory", *Philosophy of Science* 40: 194-212.
- . (1993), *Physics and Chance: Philosophical Issues in the Foundations of Statistical Mechanics*. Cambridge: Cambridge University Press.
- Truesdell, Clifford (1961), "Ergodic Theory in Classical Statistical Mechanics", in P. Caldirola (ed.), *Ergodic Theories*, volume 14 of *Proceedings of the International School of Physics "Enrico Fermi"*. New York: Academic Press, pp. 21-56.
- Wightman, Arthur S. (1983), "Regular and Chaotic Motions in Dynamical Systems: Introduction to the Problems", in G. Velo and A. S. Wightman, (eds.), *Regular and Chaotic Motions in Dynamic Systems*. New York: Plenum Press, pp. 1-26.
- Wilson, Kenneth G. and J. Kogut (1974), "The Renormalization Group and the ϵ Expansion", *Physics Reports* 12: 75-200.